

## Patient Profile

Saubhik Bhaumik, 27-year-old male

## Summary of Results

Liver function tests demonstrate values entirely within reference ranges, indicating preserved hepatic synthetic and excretory capacity. **ML Result: Normal** – the predictive model assesses the profile as within normal limits.

## Detailed Test Explanations

**Serum Bilirubin (Total):** Measures combined direct and indirect bilirubin, reflecting hepatic conjugation and excretion; elevations suggest hepatocellular injury, biliary obstruction, or hemolysis. Patient result: 0.9 mg/dL (reference: 0.2-1.2 mg/dL) – normal.

**Serum Bilirubin (Direct):** Assesses conjugated bilirubin; increased levels indicate biliary obstruction or intrahepatic cholestasis. Patient result: 0.2 mg/dL (reference: 0-0.3 mg/dL) – normal.

**Serum Bilirubin (Indirect):** Reflects unconjugated bilirubin from heme breakdown; elevations occur in hemolysis or Gilbert syndrome. Patient result: 0.70 mg/dL (reference: 0.2-1.0 mg/dL) – normal.

**SGPT (ALT):** Alanine aminotransferase, a sensitive marker of hepatocellular injury; elevations signify acute hepatitis or toxin exposure. Patient result: 36 U/L (reference: 13-40 U/L) – normal.

**SGOT (AST):** Aspartate aminotransferase, present in liver and muscle; rises indicate hepatocellular damage, though less specific than ALT. Patient result: 32 U/L (reference: 0-37 U/L) – normal.

**Serum Alkaline Phosphatase:** Enzyme elevated in cholestasis, bone disease, or biliary obstruction; not reported in this panel.

**Serum Protein:** Total protein evaluates nutritional status and hepatic synthesis; hypoalbuminemia or hypoproteinemia suggests chronic liver disease. Patient result: 7.2 g/dL (reference: 6.4-8.3 g/dL) – normal.

**Serum Albumin:** Primary protein synthesized by hepatocytes, reflecting chronic liver function and nutritional adequacy; low levels indicate synthetic dysfunction. Patient result: 4.7 g/dL (reference: 3.5-5.2 g/dL) – normal.

**Globulin:** Immunoglobulin fraction; alterations may signal inflammation or dysproteinemia. Patient result: 2.50 g/dL (reference: 1.8-3.6 g/dL) – normal.

**A/G Ratio:** Albumin-to-globulin ratio; reversal suggests chronic liver disease or protein loss. Patient result: 1.88 (reference: 1.1-2.4) – normal.

## Overall Interpretation

Comprehensive liver function panel reveals no biochemical evidence of hepatic dysfunction, inflammation, cholestasis, or synthetic impairment. All parameters fall within physiologic norms, corroborated by **ML Result: Normal**, supporting intact liver health without predictive anomalies.

## Recommendations

Maintain balanced nutrition, limit alcohol intake, engage in regular physical activity, and avoid hepatotoxins. Routine annual hepatic screening advised for ongoing monitoring, with clinical correlation if symptoms arise.

## Disclaimer

This report is for informational purposes only and does not constitute medical advice. Consult a qualified healthcare provider for diagnosis and management.